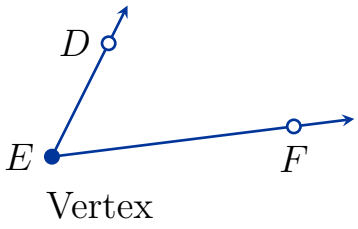
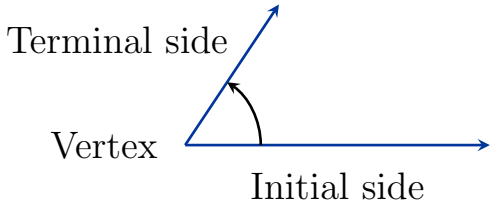
5.1: Angles

Definition.
A **ray** consists of one point on a line, and all points extending in one direction from that point. The following is ray $\overrightarrow{EF}$:

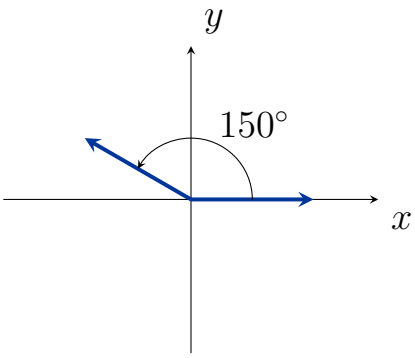


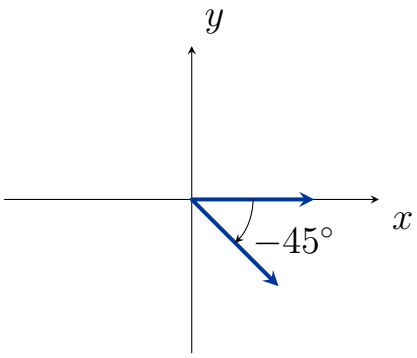
An **angle** is the union of two rays having a common endpoint. The endpoint is called the **vertex** of the angle. Angles can be named using a point on each ray and the vertex. The following is $\angle DEF$:





When we create an angle, we fix one ray, called the **initial side**, and rotate the **terminal side** about the vertex.
An angle drawn on the xy-plane is said to be in **standard position** if its vertex is located at the origin, and the initial side extends along the x-axis:

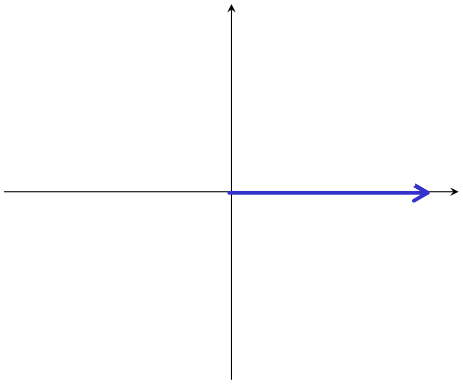




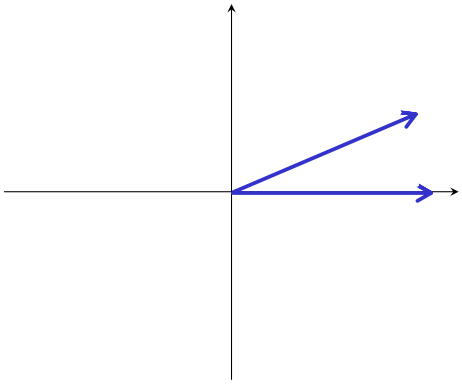
A **positive angle** rotates in the counterclockwise direction. A **negative angle** rotates in the clockwise direction. A full circle rotation is defined to be 360°.

Example. Sketch the following angles in standard position:

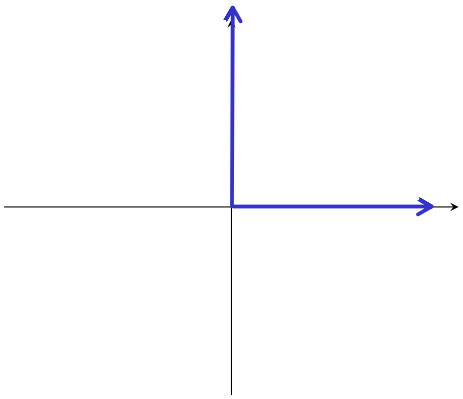
$0°$



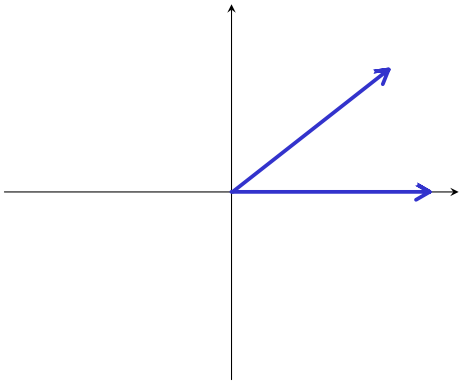
$30°$



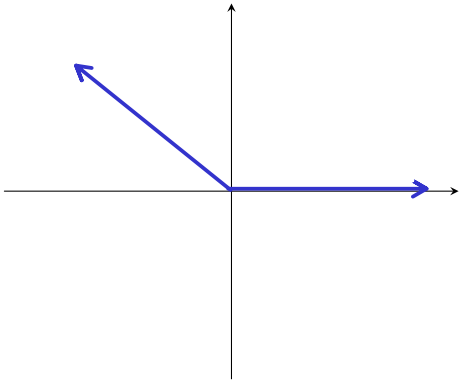
$90°$



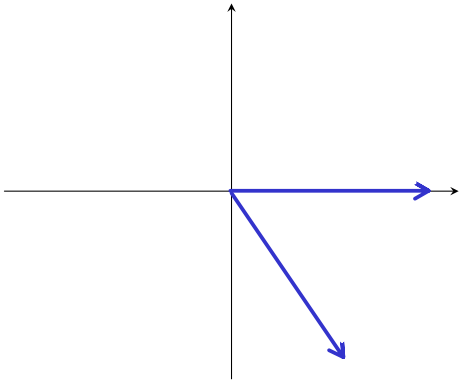
$45°$



$135°$



$-60°$



Definition.
A **radian** is the measure of a central angle corresponding to an arc equal in length to the radius of that circle. Since the circumference of a circle is $C = 2\pi r$, there are 2π radians in a full circle. This gives us

$$2\pi = 360^\circ$$

or, equivalently,

$$\pi = 180^\circ$$

Example. Convert the following angles to radians:

$225^\circ \quad \frac{\pi}{180^\circ} = \frac{5\pi}{4} = \frac{5}{4}\pi$

$270^\circ \quad \frac{\pi}{180^\circ} = \frac{3\pi}{2}$

$-135^\circ \quad \frac{\pi}{180^\circ} = -\frac{3\pi}{4}$

$-45^\circ \quad \frac{\pi}{180^\circ} = -\frac{\pi}{4}$

Example. Convert the following angles to degrees:

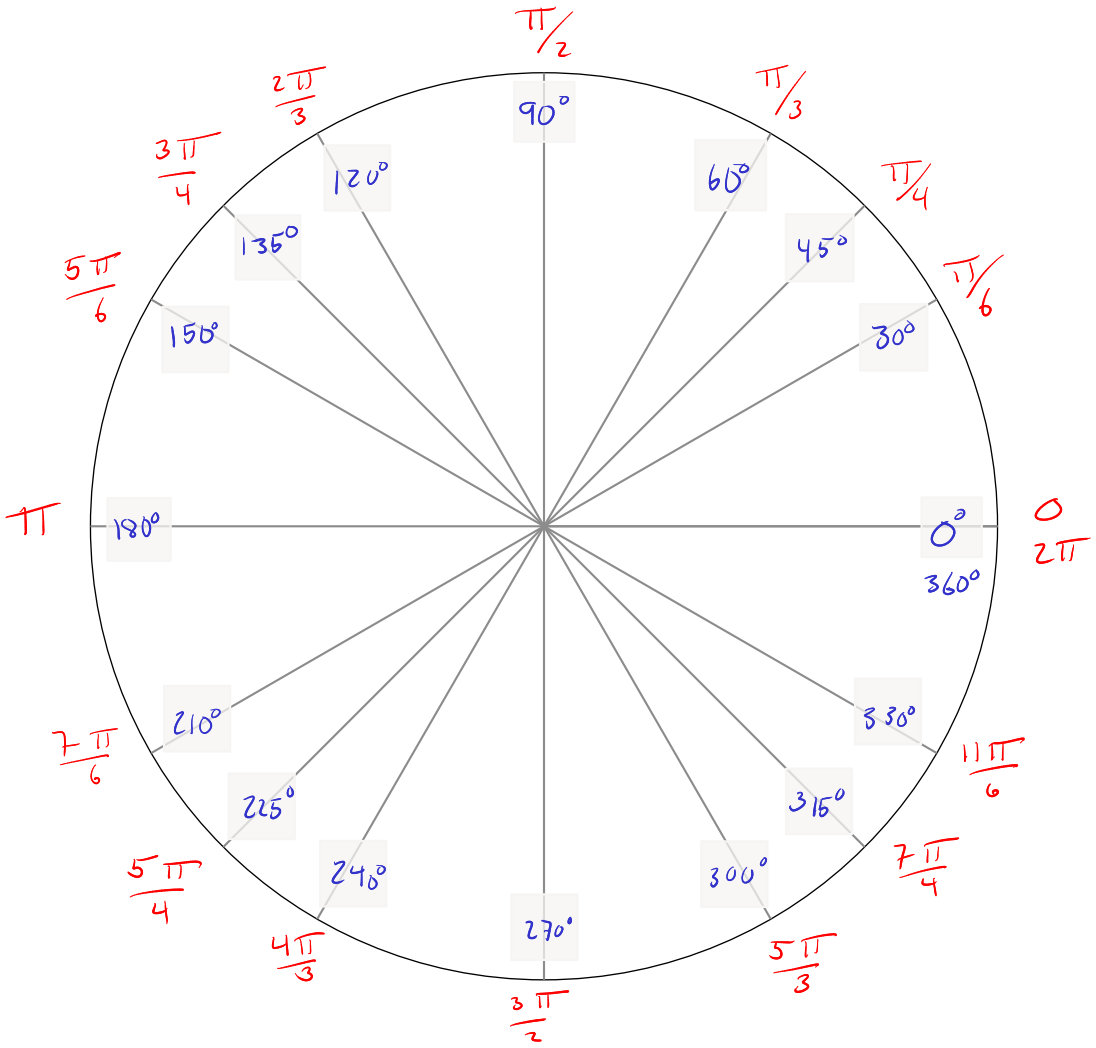
$4\pi \quad \frac{180^\circ}{\pi} = 720^\circ$

$-\frac{\pi}{2} \quad \frac{180^\circ}{\pi} = -90^\circ$

$\frac{\pi}{12} \quad \frac{180^\circ}{\pi} = 15^\circ$

$-\frac{5\pi}{4} \quad \frac{180^\circ}{\pi} = -225^\circ$

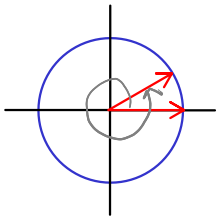
Example. The following circle contains certain angles of interest. Find the angles of each in both degrees and radians:



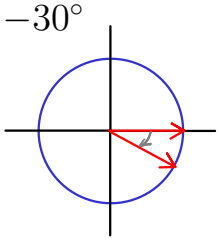
Definition.
Coterminal angles are two angles in standard position that have the same terminal side.

Example. Find a coterminal angle for each of the following:

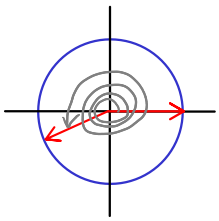
$400°$ $400° - 360° = \boxed{40°}$



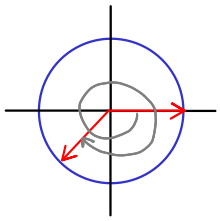
$-30°$ $-30° + 360° = \boxed{330°}$



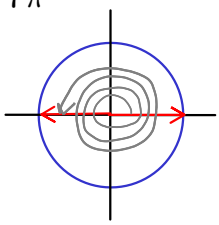
$1655°$ $1655° - 4(360°) = \boxed{215°}$



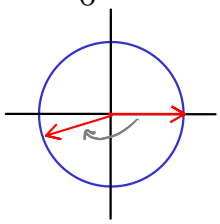
$-500°$ $-500° + 360° = \boxed{140°}$



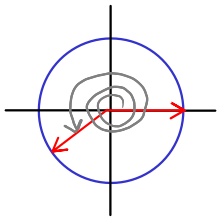
7π $7\pi - 3(2\pi) = \boxed{\pi}$



$\frac{-5\pi}{6}$ $\frac{-5\pi}{6} + 2\pi = \boxed{\frac{7\pi}{6}}$



$\frac{21\pi}{4}$ $\frac{21\pi}{4} - 2(2\pi) = \boxed{\frac{5\pi}{4}}$



$-\frac{17\pi}{3}$ $-\frac{17\pi}{3} + 3(2\pi) = \boxed{\frac{\pi}{3}}$

